

LOUNI ANYA

AI / Web Developer · Fullstack Engineer · Applied Machine Learning Research

Alger, Algérie • anyal.louni@gmail.com • +213 563 227 447 • [linkedin.com/in/anya-louni-93337b332](https://www.linkedin.com/in/anya-louni-93337b332) • github.com/Any-Louni

PROFILE

Fullstack developer and applied ML/AI researcher with a dual background in Embedded Systems & AI (B.Sc.) and Geomatics / Spatial Data (A.S.). Builds end-to-end web applications with Next.js and React, and ships rigorously benchmarked ML research with public writeups and live demos rather than notebooks alone, spanning computer vision, sensor signal processing, and workforce optimization. Trilingual (English · French DALF C2 · Arabic), with experience delivering technical projects in multicultural, cross-border environments.

EXPERIENCE

STEAM Mentor | *World Learning Algeria* | June 2026 – Present

- Certified STEAM Mentor delivering hands-on Python and machine learning workshops to student audiences.
- Applied structured, rules-based instructional design to make technical concepts approachable for beginner learners.
- Built and delivered supporting workshop material, including a Linux fundamentals session for a teacher training track.

Development Co-Manager | *NIT Computer Society, Alger* | Oct 2025 – Dec 2025

- Designed and built internal dashboards to track participant registrations across club events and programs.
- Supported HR processes with lightweight custom tooling to replace manual tracking.
- Coordinated technical work across a small team, from initial scoping through delivery.

Design Co-Manager | *NIT Computer Society, Alger* | Oct 2024 – Sep 2025

- Designed the club's logo and a branded notebook/planner, applying consistent visual identity guidelines.
- Followed established design principles (typography, color, layout) to keep materials cohesive across formats.
- Coordinated with the development team to align visual and technical deliverables on shared timelines.

International Peer Leader, OISS | *Portland Community College, Oregon USA* | Aug 2023 – Aug 2024

- Created bilingual (EN/FR) digital resources and communications reaching students from 30+ countries.
- Coordinated multi-stakeholder international programs at the OISS.
- Gained firsthand context on Pacific Northwest coastal tech communities.

PROJECTS

Full technical writeups, code, and live demos: github.com/Any-Louni

Tawzia, AI-Powered Workforce & HR Platform for Algeria | *AI Forge Program, Boeing-sponsored via World Learning*

- Built an Algeria-specific workforce engine covering the full employee lifecycle: hiring, scheduling, payroll, compliance, and retention, encoding the Hijri calendar and Algerian labor law (SNMG, CNAS, IRG) directly into the system logic.
- Combined scikit-learn demand forecasting with linear-programming roster optimization (scipy.linprog) to generate staffing schedules under real compliance constraints, plus attrition-risk and CV-screening modules.
- Added an aviation delay-cascade model projecting downstream workforce impacts for Air Algérie / EGSA operations, and validated the concept through direct outreach and discovery calls with Algerian institutions.

Deep-Sea-OOD, Open-Set Recognition for Deep-Sea Gelatinous Zooplankton | *Independent ML research, full paper + live Streamlit demo + web app*

- Benchmarked four out-of-distribution detection methods (MSP, Energy, Mahalanobis, ViM) on real ROV footage from FathomNet, across 230 species under a 73.8x class imbalance, stratifying novel species by taxonomic distance and stress-testing every method against synthetic low-light, blur, scale-loss, and compression degradation.
- Found that the best clean-data method (ViM, AUROC 0.677) is also the least robust under degradation, while Mahalanobis distance, the weaker clean-data performer, becomes the most reliable choice once conditions degrade: a reversal with direct implications for deploying OOD detection on real underwater cameras rather than curated benchmarks.
- Root-caused all 16 zero-F1 classes to specific, named failure mechanisms (split artifacts, single-dive domain shift, thin classes, taxonomic-label competition) rather than reporting them as unexplained noise, and shipped the full analysis as a public paper, dataset card, and interactive demo.

Galaxy-Compass, Rotation-Equivariant CNNs for Galaxy Morphology | *Independent ML research, self-supervised atlas + anomaly detector + web demo*

- Built an E(2)-steerable convolutional network for Galaxy10 DECaLS morphology classification and ran a controlled, matched-baseline comparison against a dense CNN with identical channel widths, augmentation, and compute, isolating the effect of encoding rotational symmetry directly into the architecture.
- Demonstrated a ~7x label-efficiency gain (statistically confirmed: 15/15 paired comparisons favor the steerable model, sign-test $p = 6.1 \times 10^{-5}$) while using 11.8x fewer parameters, and verified true equivariance to 10^{-7} precision, versus a 2.6×10^{-2} drift for the standard CNN.
- Extended the encoder into a self-supervised SimCLR atlas over all 17,736 galaxies and a two-signal anomaly ranker (VAE reconstruction error + embedding-space isolation) that enriches for mergers and complex spirals by up to 1.7x, shipped as a public, GPU-free, laptop-reproducible repository with an interactive results site.

EDUCATION

B.Sc. Embedded Systems & Artificial Intelligence | *Numidia Institute of Technology, Alger, Algérie* | 2024 – 2027 (Expected)

A.S. Geographic Information Science & Cartography (GIS) | *Portland Community College, Oregon, USA* | Apr 2023 – Aug 2024

PROGRAMS & CERTIFICATIONS

- GCI World 2026 (Matsuo Lab, University of Tokyo): ongoing, accepted participant, starting September 2026
- Thirduni Mentorship & Career Development Program: ongoing, accepted participant
- AI Forge, Boeing-sponsored program, World Learning
- Certified STEAM Mentor, World Learning
- Google Data Analytics, SQL, Data Analysis (2025)
- Scientific Computing in Python, freeCodeCamp (2025)
- Deep Learning, NVIDIA (2026)
- Intro to Cybersecurity, Cisco (2025)
- DALF C2 French, highest level, native-equivalent
- Career Essentials in Cybersecurity, Microsoft × LinkedIn (2025)
- TESOL Certified

TECHNICAL SKILLS

Frontend

Next.js	React	TypeScript	Tailwind CSS
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Backend

Node.js	REST APIs	SQL / DBMS
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AI / Data

Python	PyTorch	scikit-learn	Streamlit
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Tools

Git / GitHub	Linux	Cybersecurity Basics
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LANGUAGES

English (Fluent) · **French** (DALF C2, bilingual) · **Arabic** (Native)